Finding report: Polynomial calculus degree above p for the onto functional pigeonhole principle with surplus divisible by p squared

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1 Introduction

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Setting

The onto functional pigeonhole principle onto-FPHP$_n^{n+k}$ says that $n+k$ pigeons can be mapped one-to-one onto n holes; it is false for every $k \geq 1$. In Boolean variables x_{ij} (pigeon i in hole j) its axioms are $\sum_j x_{ij} - 1$, $\sum_i x_{ij} - 1$, $x_{ij}x_{i'j}$ and $x_{ij}x_{ij'}$, the linear encoding used for this family [1, 2]. These results are for the linear encoding only: translating the CNF clause by clause instead turns each pigeon clause into a polynomial of degree n [2], so degree bounds below n say nothing about that encoding. A polynomial calculus (PC) refutation over $\mathbb{F}_p$ derives 1 from the axioms and the Boolean axioms $x^2 - x$ by linear combinations and multiplication by a variable; its degree is the largest degree of a line.

When p does not divide k there is a degree-1 refutation. When p divides k but p^2 does not, the PC degree is exactly p for $n \geq 2p-3$ (a companion finding on this site). This finding is about surplus divisible by p^2.

Result

For every prime p, every $c \geq 1$, and every $n \geq 2p-1$, onto-FPHP$_n^{n+cp^2}$ has no PC refutation over $\mathbb{F}_p$ of degree at most p. Its PC degree is therefore at least $p+1$. With the companion result, surplus divisible by p^2 is strictly harder for PC over $\mathbb{F}_p$ than surplus divisible only by p, for every prime and from a linear threshold in n.

The proof constructs, for every prime p, an explicit integer matrix on $2p-1$ rows and columns (a "square lift"). From it comes a balanced design, which serves as a degree-p witness that no refutation of that degree exists. The cases

$p = 2$ ($n \geq 3$, surplus $4c$) and $p = 3$ ($n \geq 5$, surplus $9c$) are also proved separately.

Why it matters

Razborov's Question 7 asks whether PC degree for surplus p^ℓ stays bounded in characteristic p [2]. At $\ell = 2$ the known PC lower bound is 3, for $N \geq 2(21 + 3p^2) - p$ [4]. This result gives $p + 1$ from $n = 2p - 1$: larger for every $p \geq 3$, and at a far smaller n. Earlier runs of this programme proved $p + 1$ for $p = 3, 5, 7$ separately; this result covers every prime at once.

Proved vs checked

Everything is proved in Lean 4 and kernel-checked, for all primes p, all c and all $n \geq 2p - 1$; nothing is checked only up to a bound. The proofs use only the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The main theorem is stated for standard (syntactic) PC with Boolean axioms. A second agent in the run re-proved all six statements in Lean, but its re-proofs reuse six of the author's lemmas each, so they are not fully independent. No person has reviewed the proofs; the evidence is the kernel check.

Novelty statement

This result extends:

- Riis-Sitharam, ECCC TR97-048, Theorem 33: PC degree >= 2^l - 1 for onto-PHP^{N+p^l}_N over characteristic p once N >= (21+3p^2)(2^l - 2) - p (same family; proof not checked by us; at l = 2 this is >= 3 against our >= p+1 from n = 2p-1; at p = 2, l = 3 it is >= 7 for N >= 196 against our >= 5 for n >= 7).
- Beame-Riis 1998, Theorem 8.1 and Lemma 8.7: Nullstellensatz degree between 2^l - 1 and p^l - 1 (multiplier degree) for the same family; Section 10 lists polynomial-calculus lower bounds as open.
- Riis 1993, as Theorem 8 of Razborov's survey: degree-d PC refutations when C(m,d) and C(n,d) differ mod p, which gives the matching upper bound p for k = pc, p not dividing c (via Lucas).
- Razborov, Proof complexity of pigeonhole principles (2001), Question 7: the fixed-p, fixed-l bounded-degree question for onto-FPHP^{n+p^l}_n.
- Miksa-Nordstrom (JACM 2024): general PC degree method; onto-FPHP with m = n+1 easy (Riis); the case of characteristic dividing the surplus not covered.
- BGIP 2001: proves linear PC degree for Tseitin mod p over characteristic q containing a primitive root of unity of order p (Thm 15) and for

MOD_p^n over F_q, q not dividing p (Cor 20: "requires degree > delta n"). This is the cross-characteristic regime. Our results concern the same-characteristic regime p | k, so there is no overlap. The paper does not treat onto-PHP.

- Alekhnovich-Razborov 2003: "immunity" method giving linear PC degree bounds for mod_p Tseitin over characteristic different from p, for flow tautologies and for random k-CNF in characteristic 2 (abstract only). No PHP or onto-PHP result. The method is relevant only as possible technique.

- Impagliazzo-Pudlak-Sgall 1999: simpler proof of Razborov's ceil(n/2)+1 PC lower bound for (injective, non-onto) PHP^m_n over any field, plus an upper bound in large characteristic (abstract). This is a different principle: without the onto and functional axioms the bound is independent of p. It does not transfer to the exact degree p or p+1 for onto-FPHP.

- BIKPRS 1996/97: Nullstellensatz lower bounds for Count_q over Z_m with q not dividing m (Thm 3.2) and over Q (Thm 4.6), plus the PC-to-NS simulation (Thm 5.4). This is Nullstellensatz and Count, not PC degree for onto-FPHP, and the paper excludes the case q | m.

- Grigoriev FOCS 1998 (snippet only, not opened): Nullstellensatz degree for Tseitin mod p over other characteristics. It is a predecessor of BGIP; NS, not PC, and a different principle.

- Filmus-Lauria-Mika-Nordström-Vinyals 2025: PC space and separations. It mentions the functional PHP only for space limits, not for characteristic-dependent degree.

- Ken 2022/2024: restates Razborov's injPHP bound "no PC-refutation ... with degree <= m/2" (Thm 24) and extends it to NS over finite rings (Cor 25). Onto-PHP appears only in reductions (Props 12, 15), and no degree bound for it is proved.

- Reduction transfer: none of the opened sources gives a low-degree reduction from Count_q, Tseitin mod q or Mod_q (q != p) to onto-FPHP^{n+k}_n with p | k over F_p. Since the cross-characteristic lower bounds need q != p, they do not yield results 1-3 directly.

Our prior-art search (2026-11-06) covered:

- Riis, Sitharam, Non-constant degree lower bounds imply linear degree lower bounds, ECCC TR97-048 (1997), pp. 36-38 and 45-46 (Theorem 33, Propositions 27 and 28), opened 2026-10-29

- Beame, Riis, More on the relative strength of counting principles, DIMACS Series 39 (1998), homes.cs.washington.edu/~beame/papers/riis.pdf: Theorem 8.1, Lemma 8.7, Section 10, opened 2026-10-27 and 2026-11-01

- Razborov, Proof complexity of pigeonhole principles, DLT 2001 (LNCS 2295), people.cs.uchicago.edu/~razborov/files/php_survey.pdf: Theorem 8, Question 7, footnote 5, opened 2026-10-27 and 2026-10-29
- Riis, BRICS Report RS-99-29, opened 2026-10-29
- Miksa, Nordstrom, A generalized method for proving polynomial calculus degree lower bounds (CCC 2015; JACM 2024; arXiv 1505.01358), opened 2026-10-27
- de Rezende et al., Theory of Computing 21(6) (2025), opened 2026-10-29
- Lauria et al., Theory of Computing 21(4) (2025), opened 2026-10-27
- Buss, Grigoriev, Impagliazzo, Pitassi, "Linear gaps between degrees for the polynomial calculus modulo distinct primes", JCSS 62(2):267-289, 2001 (STOC 1999). Opened: journal final PDF, https://mathweb.ucsd.edu/~sbuss/ResearchWeb/lineargaps/j
- Alekhnovich, Razborov, "Lower bounds for polynomial calculus: non-binomial case", Proc. Steklov Inst. Math. 242 (2003) 18-35 (FOCS 2001). Opened: abstract only, https://mathnet.ru/eng/tm403 (full text not opened).
- Impagliazzo, Pudlak, Sgall, "Lower bounds for the polynomial calculus and the Groebner basis algorithm", Computational Complexity 8(2):127-144, 1999 (ECCC TR97-042). Opened: ECCC abstract only, https://eccc.weizmann.ac.il/report/1997/04
- Buss, Impagliazzo, Krajicek, Pudlak, Razborov, Sgall, "Proof complexity in algebraic systems and bounded depth Frege systems with modular counting", Computational Complexity 6(3):256-298, 1996/97. Opened: full PDF, https://mathweb.ucsd.edu/~sbuss/ResearchWeb/nullstellensatz/token.pdf
- Grigoriev, "Tseitin's tautologies and lower bounds for Nullstellensatz proofs", FOCS 1998, pp. 648-652. Snippet only, not opened (dblp/researchr entries).
- Mika, Nordström, "A generalized method for proving polynomial calculus degree lower bounds", CCC 2015 (arXiv 1505.01358). Already credited. Seen in search results; not re-opened here.
- Filmus, Lauria, Mika, Nordström, Vinyals, "Towards an understanding of polynomial calculus: new separations and lower bounds", Theory of Computing 21(4), 2025 (ICALP 2013). Opened: abstract page, https://zesty2.cs.uchicago.edu/articles/v021a0
- Eitetsu Ken, "On some Sigma_0^B-formulae generalizing counting principles over V^0", arXiv 2203.10237 (2022; a journal version appears to be Arch. Math. Logic 2024, https://link.springer.com/article/10.1007/s00153-024-00938-1). Opened: ar5iv HTML and Springer page, first ~100k characters of each.

- Web searches for 2000-2026 papers on PC degree of onto-PHP or Count where the characteristic divides the surplus: no further hits beyond the items above.

Prior-art answer recorded by the owner (basis stated in the file); published in `prior-art/`.

If you know of related work we missed, please open an issue.

This result was found by AI research agents (the autoresearch harness). It was proved in Lean; the verification steps below build the published programme library.

How it was produced

The declaring Lean sources are in `checks/library/`.

How to verify

Lean toolchain: `leanprover/lean4:v4.35.0`. `https://github.com/kodakodapa/autoresearch-findings/tree/2abad0979eeb03ede3a77490f45a4f381119ff02/programmes/proof-complexity`. Install elan and open a new shell after installing elan, then pin the library subtree while keeping the finding checks:

```
git clone https://github.com/kodakodapa/autoresearch-findings.git
cd autoresearch-findings
git checkout 2abad0979eeb03ede3a77490f45a4f381119ff02 -- programmes/proof-complexity
cd programmes/proof-complexity && lake build 'Programme.ń20261102-1249-proof-complexityż.Uni
AR_LIBRARY=$PWD python -I ../../claims/pc-degree-above-p-onto-fphp-surplus-p-squared/checks/
AR_LIBRARY=$PWD python -I ../../claims/pc-degree-above-p-onto-fphp-surplus-p-squared/checks/
AR_LIBRARY=$PWD python -I ../../claims/pc-degree-above-p-onto-fphp-surplus-p-squared/checks/
AR_LIBRARY=$PWD python -I ../../claims/pc-degree-above-p-onto-fphp-surplus-p-squared/checks/
AR_LIBRARY=$PWD python -I ../../claims/pc-degree-above-p-onto-fphp-surplus-p-squared/checks/
AR_LIBRARY=$PWD python -I ../../claims/pc-degree-above-p-onto-fphp-surplus-p-squared/checks/
```

A full `lake build` builds the whole library. The named targets build only the claim modules and their dependencies. Indicative measurements from the 2026-11-06 reproduction on 2 cores / 7.8 GB: K13–K15 90 s; K1–K6 440 s; K18 87 min (90 certificate modules). Peak memory was 4.7 GB for one Lean process. Build times depend on hardware and cached dependencies.

`checks/run-copies/`: the author's working copies from the run; they import the run's private `Work` package and are not built standalone.

2 Claims

Claim K1

For every prime p, SquareLiftAssumption(p) holds, i.e. there exists an integral square lift on $2p - 1$ rows and columns.

Status: proved (axioms: propext, Classical.choice, Quot.sound) ů second-agent re-proof (skeptic/codex).

Re-checked at publication on 2026-11-06: pass (`results/publication/lean-K1.json`).

Verified. No check scripts.

Evidence.

- library check `checks/lean-K1.py` (library theorem `Programme.ń20261102-1249-proof-complexityż.Un`
- proof `checks/run-copies/K1-Unconditional.lean`: Work.UniformC3.squareLiftAssumption — pass (`results/K1-1.json`)
- reproof `checks/reproofs/K1-Reproof.lean` by the skeptic (codex): `Work.UniformC3.squareLiftAssum` — pass (`checks/reproofs/K1-Reproof.json`); second-agent re-proof; uses 6 lemmas from the author's package; re-proof by a second agent; imports the run's private packages (not rebuildable from this repo)

Prior art. Partial prior art.

Claim K2

For every prime p and natural c, some T satisfies $\mathrm{Bal}(p, 2p-1, cp^2, T, T, e)$ for every $e \le p-1$, and $p \mid T([]) - 1$.

Status: proved (axioms: propext, Classical.choice, Quot.sound) ů second-agent re-proof (skeptic/codex).

Re-checked at publication on 2026-11-06: pass (`results/publication/lean-K2.json`).

Verified. No check scripts.

Evidence.

- library check `checks/lean-K2.py` (library theorem `Programme.ń20261102-1249-proof-complexityż.Un`
- proof `checks/run-copies/K1-Unconditional.lean`: Work.UniformC3.uniform_unit_design_uncon — pass (`results/K2-1.json`)
- reproof `checks/reproofs/K2-Reproof.lean` by the skeptic (codex): `Work.UniformC3.uniform_unit_de` — pass (`checks/reproofs/K2-Reproof.json`); second-agent re-proof; uses 6 lemmas from the author's package; re-proof by a second agent; imports the run's private packages (not rebuildable from this repo)

Prior art. Partial prior art.

Claim K3

For every prime p, natural c, and $n \geq 2p-1$, ontoFPHP(n, cp^2) has no degree-p PPCRefutation modulo p (functional encoding, PC line degree).

Status: proved (axioms: propext, Classical.choice, Quot.sound) ů second-agent re-proof (skeptic/codex).

Re-checked at publication on 2026-11-06: pass (`results/publication/lean-K3.json`).

Verified. No check scripts.

Evidence.

- library check `checks/lean-K3.py` (library theorem `Programme.ń20261102-1249-proof-complexityż.Un`
- proof `checks/run-copies/K1-Unconditional.lean`: Work.UniformC3.uniform_no_degree_p_unc — pass (`results/K3-1.json`)
- reproof `checks/reproofs/K3-Reproof.lean` by the skeptic (codex): `Work.UniformC3.uniform_no_degr` — pass (`checks/reproofs/K3-Reproof.json`); second-agent re-proof; uses 6 lemmas from the author's package; re-proof by a second agent; imports the run's private packages (not rebuildable from this repo)

Prior art. Partial prior art.

Claim K4

For every prime p, natural c, and $n \geq 2p-1$, ontoFPHP(n, cp^2) has no standard polynomial-calculus refutation modulo p of line degree at most p (functional encoding).

Status: proved (axioms: propext, Classical.choice, Quot.sound) ů second-agent re-proof (skeptic/codex).

Re-checked at publication on 2026-11-06: pass (`results/publication/lean-K4.json`).

Verified. No check scripts.

Evidence.

- library check `checks/lean-K4.py` (library theorem `Programme.ń20261102-1249-proof-complexityż.Un`
- proof `checks/run-copies/K1-Unconditional.lean`: Work.UniformC3.uniform_no_standard_degr — pass (`results/K4-1.json`)
- reproof `checks/reproofs/K4-Reproof.lean` by the skeptic (codex): `Work.UniformC3.uniform_no_star` — pass (`checks/reproofs/K4-Reproof.json`); second-agent re-proof; uses 6 lemmas from the author's package; re-proof by a second agent; imports the run's private packages (not rebuildable from this repo)

Prior art. Partial prior art.

Claim K5

For all natural c and $n \geq 3$, ontoFPHP$(n, 4c)$ has no standard polynomial-calculus refutation modulo 2 of line degree at most 2 (functional encoding).

Status: proved (axioms: propext, Classical.choice, Quot.sound) ů second-agent re-proof (skeptic/codex).

Re-checked at publication on 2026-11-06: pass (`results/publication/lean-K5.json`).

Verified. No check scripts.

Evidence.

- library check `checks/lean-K5.py` (library theorem `Programme.ń20261102-1249-proof-complexityż.Un`
- proof `checks/run-copies/K5-Main.lean`: Work.UniformC3.no_standard_degree_two — pass (`results/K5-1.json`)
- reproof `checks/reproofs/K5-Reproof.lean` by the skeptic (codex): `Work.UniformC3.no_standard_deg` — pass (`checks/reproofs/K5-Reproof.json`); second-agent re-proof; uses 6 lemmas from the author's package; re-proof by a second agent; imports the run's private packages (not rebuildable from this repo)

Prior art. Partial prior art.

Claim K6

For all natural c and $n \geq 5$, ontoFPHP$(n, 9c)$ has no standard polynomial-calculus refutation modulo 3 of line degree at most 3 (functional encoding).

Status: proved (axioms: propext, Classical.choice, Quot.sound) ů second-agent re-proof (skeptic/codex).

Re-checked at publication on 2026-11-06: pass (`results/publication/lean-K6.json`).

Verified. No check scripts.

Evidence.

- library check `checks/lean-K6.py` (library theorem `Programme.ń20261102-1249-proof-complexityż.Un`
- proof `checks/run-copies/K5-Main.lean`: Work.UniformC3.no_standard_degree_three — pass (`results/K6-1.json`)
- reproof `checks/reproofs/K6-Reproof.lean` by the skeptic (codex): `Work.UniformC3.no_standard_deg` — pass (`checks/reproofs/K6-Reproof.json`); second-agent re-proof; uses 6 lemmas from the author's package; re-proof by a second agent; imports the run's private packages (not rebuildable from this repo)

Prior art. Partial prior art.

Provenance. Run `20261102-1249`, claims K1, K2, K3, K4, K5, K6, run-folder commit `359fa20477b8d26a5d2ef8a16e5465474e74b452`, published 2026-11-06.

3 Discussion

This is a lower bound, not an exact value. The best upper bound we know at surplus p^2 is p^2, from Riis's criterion [3, 2]. So for surplus p^2 the degree lies between $p+1$ and p^2. At $p=2$ the separate exact result that the degree is 4 for surplus 4 ($n \geq 5$) is in the same library. The bound does not grow with ℓ and does not answer Question 7. We have not checked the proof of Riis and Sitharam's Theorem 33, which we cite for comparison only.

4 How to verify

For programme findings, follow the clone, build and run steps in each finding. The declaring Lean sources are in `checks/library/`; the publication checks import their modules and print each theorem's axioms. The output of the checks re-run at publication is in `claims/<slug>/results/publication/`. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

5 Revisions

- 2026-11-06 — Math rendering on the site; paper lists only cited references
- 2026-11-06 — Library published buildable; verification steps; run copies labelled
- 2026-11-06 — Verification steps; stale run copies removed; encoding stated
- 2026-11-06 — Publication re-checks and reproofs published; prior art extended; texts corrected
- 2026-11-06 — Library theorem names and checked statements shown; reproof wording corrected; build cost

References

[1] Paul Beame and Søren Riis. More on the relative strength of counting principles. In *Proof Complexity and Feasible Arithmetics*, volume 39 of *DIMACS Series in Discrete Mathematics and Theoretical Computer Science*, pages 13–35. American Mathematical Society, 1998.

[2] Alexander A. Razborov. Proof complexity of pigeonhole principles. In *Developments in Language Theory (DLT 2001)*, volume 2295 of *Lecture Notes in Computer Science*, pages 100–116. Springer, 2002.

[3] Søren Riis. *Independence in Bounded Arithmetic*. PhD thesis, University of Oxford, 1993.

[4] Søren Riis and Meera Sitharam. Non-constant degree lower bounds imply linear degree lower bounds. Technical Report TR97-048, Electronic Colloquium on Computational Complexity, 1997.